Moed A Solution

1

1.1

50% of the energy is transmitted through the polarizer: $I_A = \frac{I_0}{2}$. The light is polarized in the $\hat{x}$ direction.

1.2

If the middle polarizer is removed, no light is transmitted through the third polarizer. This is because the light reaching the third polarizer is polarized in the $\hat{x}$ direction while the polarizer is oriented in the $\hat{y}$ direction.

1.3

Only the component that is parallel to the polarizer is transmitted, i.e. the projection of the field onto the axis of the polarizer.

$$\vec{E}_{transmitted} = E\left(\hat{x}\cdot\hat{e}_\theta\right)\hat{e}_\theta = E\hat{x}\cdot\begin{pmatrix}\cos\left(\theta\right)\\ \sin\left(\theta\right)\end{pmatrix} = E\cos\left(\theta\right)\hat{e}_\theta$$

where $\hat{e}_\theta = \begin{pmatrix}\cos\left(\theta\right)\\ \sin\left(\theta\right)\end{pmatrix}$.
The transmitted intensity:

$$I_B = I_A\cos^2\left(\theta\right)$$

and the field is polarized in the direction of the polarizer: $\hat{e}_\theta = \begin{pmatrix}\cos\left(\theta\right)\\ \sin\left(\theta\right)\end{pmatrix}$

1.4

Again, we need to find the projection onto the direction of the polarizer.

$$I_c = I_B \sin(\theta) = \frac{I_0}{2} \cos^2(\theta) \sin^2(\theta)$$

We find that introducing a polarizer in between two perpendicular polarizers can increase the transmitted intensity, even though $I_B < I_A$! This occurs because the field after polarizer B has a component in the $\hat{y}$ direction, as oppose to the case without this polarizer. This component is maximized when $\theta = \frac{\pi}{4}$.

1.5

We begin with $N = 2$, i.e. two polarizers in the middle $\delta\theta = \frac{\pi}{6}$, and then generalize to any N.
After first polarizer ($\hat{x}$ direction):

$$\vec{E}_1 = \frac{E_0}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

After second:

$$\vec{E}_2 = \left(\vec{E}_1 \cdot \hat{e}_2\right) \hat{e}_2$$

$$\hat{e}_2 = \begin{pmatrix} \cos\left(\frac{\pi}{6}\right) \\ \sin\left(\frac{\pi}{6}\right) \end{pmatrix}$$

$$\hat{E}_1 \cdot \hat{e}_2 = \cos(\theta_{12}) = \cos\left(\frac{\pi}{6}\right)$$

$$\Downarrow$$

$$\vec{E}_2 = \frac{E_0}{2} \cos\left(\frac{\pi}{6}\right) \hat{e}_2$$

After third:

$$\vec{E}_3 = \frac{E_0}{2} \cos^2\left(\frac{\pi}{6}\right) \hat{e}_3$$

$$\hat{e}_3 = \begin{pmatrix} \cos\left(\frac{\pi}{6}\right) \\ \sin\left(\frac{\pi}{6}\right) \end{pmatrix}$$

After fourth:

$$\vec{E}_4 = \frac{E_0}{2} \cos^3\left(\frac{\pi}{6}\right) \hat{e}_4$$

N polarizers in the middle:

$$\vec{E}_{transmitted} = \frac{E_0}{2}\cos^{N+1}\left(\frac{\pi/2}{N+1}\right)\hat{y}$$

$$\lim_{N\to\infty}\vec{E} = \lim_{N\to\infty}\frac{E_0}{2}\left(1-\frac{1}{2}\left(\frac{\pi/2}{N+1}\right)^2\right)^{N+1}\hat{y} = \frac{E_0}{2}$$

$$I = I_A = \frac{I_0}{2}$$

2

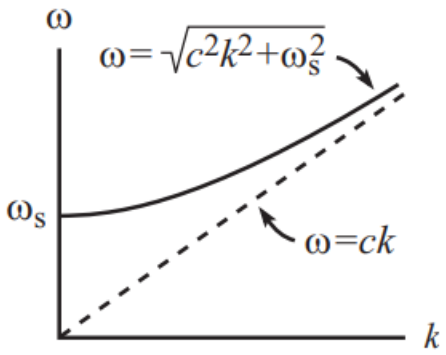
2.1

From notes:

$$\omega^2 = c^2k^2 + \omega_s^2$$

$$\omega_s^2 = \frac{\sigma}{\mu}$$

2.2



Cut-off frequency:

$$\omega_s = \sqrt{\frac{\sigma}{\mu_2}}$$

2.3

Impedance:

$$Z \equiv \frac{T}{v}$$

Dispersionless media (first media):

$$Z_1(k) = \frac{T}{v_1} = \frac{Tk_1}{\omega}$$

Second media:

$$Z_2(k) = \frac{T}{v_2} = \frac{Tk_2}{\omega}$$

We need to find the wave-number in the second media:

$$\omega_1(k_1) = \omega_1(k_2)$$

$$\Downarrow$$

$$c_1 k_1 = \sqrt{c_2^2 k_2^2 + \omega_s^2}$$

$$\Downarrow$$

$$k_2 = \frac{1}{c_2}\sqrt{c_1^2 k_1^2 - \omega_s^2}$$

Reflection:

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2} = \frac{\frac{Tk_1}{\omega} - \frac{Tk_2}{\omega}}{\frac{Tk_1}{\omega} + \frac{Tk_2}{\omega}} = \frac{k_1 - k_2}{k_1 + k_2}$$

$$= \frac{c_2 k_1 - \sqrt{c_1^2 k_1^2 - \omega_s^2}}{c_2 k_1 + \sqrt{c_1^2 k_1^2 - \omega_s^2}} = \frac{\sqrt{\frac{T}{\mu_2}} k_1 - \sqrt{\frac{T}{\mu_1} k_1^2 - \frac{\sigma}{\mu_2}}}{\sqrt{\frac{T}{\mu_2}} k_1 + \sqrt{\frac{T}{\mu_1} k_1^2 - \frac{\sigma}{\mu_2}}}$$

Similarly:

$$t = \frac{2Z_1}{Z_1 + Z_2} = \frac{2k_1}{k_1 + k_2}$$

2.4

Impedance matching:

$$Z_1 = Z_2$$

$$\Downarrow$$

$$k_1 = k_2$$

$$\Downarrow$$

$$c_2^2 k_1^2 = c_1^2 k_1^2 - \omega_s^2$$

$$\Downarrow$$

$$\frac{T}{\mu_2} k_1^2 = \frac{T}{\mu_1} k_1^2 - \frac{\sigma}{\mu_2}$$

$$\Downarrow$$

$$\frac{1}{\mu_2}\left(T k_1^2 + \sigma\right) = \frac{T}{\mu_1} k_1^2$$

$$\boxed{\mu_2 = \mu_1 \left(1 + \frac{\sigma}{T k_1^2}\right)}$$

Sanity check #1: $\mu_2 \underset{\sigma \to 0}{\to} \mu_1$

Sanity check #2: $\mu_2 \underset{k_1 \to \infty}{\to} \mu_1$, as expected from from the plot of the dispersion relations.

Maximum transmission:

When $k_1 = k_2$:

$$r = 0$$

$$t = 1$$